occurring in the real Universe.

4. Khalatnikov and Lifshitz had claimed to prove, however, that singularities are *always* unstable against small perturbations and thus never occur, and therefore the Oppenheimer–Snyder singularity could never occur in our real Universe.
5. In Princeton, at least, there was some skepticism about the Khalatnikov–Lifshitz claim. This skepticism may have been driven in part by Wheeler's desire for singularities, since they would be a "marrying" place for general relativity and quantum mechanics.

Nineteen sixty-four was a watershed year. It was the year that Roger Penrose revolutionized the mathematical tools that we use to analyze the properties of spacetime. His revolution was so important, and had such great impact on the quest for Wheeler's holy grail, that I shall digress for a few pages to describe his revolution and describe Penrose himself.

Penrose's Revolution

Roger Penrose grew up in a British medical family; his mother was a physician, his father was an eminent professor of human genetics at University College in London, and his parents wanted at least one of their four children to follow in their footsteps with a medical career. Roger's older brother Oliver was a dead loss; from an early age he was intent on a career in physics (and in fact would go on to become one of the world's leading researchers in statistical physics—the study of the behaviors of huge numbers of interacting atoms). Roger's younger brother Jonathon was also a dead loss; all he wanted to do was play chess (and in fact he would go on to become the British chess champion for seven years running). Roger's little sister Shirley was much too young, when Roger was choosing a career, to show inclinations in any direction (though she ultimately would delight her parents by becoming a physician). That left Roger as his parents' greatest hope.

At age sixteen Roger, like all the others in his class, was interviewed by the school's headmaster. It was time to decide the topics for his last two years of pre-college study. "I'd like to do mathematics, chemistry, and biology," he told the headmaster. "No. Impossible. You cannot combine